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Decentralized Hybrid Flocking Guidance for a Swarm of Small UAVs

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Abstract— **Flocking is defined as flying in groups without colliding into each other through data exchange where each UAV applies a decentralized algorithm. In this paper, hybrid flocking control is proposed by using three types of guidance methods: vector field, Cucker-Smale model, and potential field. Typically, hybrid flocking control using several methods can lead to generating conflicting commands and thus degrading the performance of the mission. To address this issue, the adaptive Cucker-Smale model is proposed. Besides, we use social learning particle swarm optimization to determine the optimal weightings between guidance methods. It is verified through numerical simulations that the optimal weighting for missions such as loitering and target tracking results in effective flocking.**

I. Introduction

In recent years, many researchers have been attracted to the efficiency and autonomy of multi-agent systems (MASs) in various fields including biology, physics, and engineering [1],[2]. The efficiency and autonomy of MASs come from cooperation between agents, which is achieved by consensus of all agents through observations or exchanges of information between agents [3]. Many studies on self-organization that mimic natural organisms are one of the representative methods for the effective operation of MASs [4]. In the operation of a large number of unmanned agents, a self-organized formation called flocking has been actively studied since it is highly applicable to many fields.

The flocking is a phenomenon when an agent satisfies certain rules among agents [5]. Cucker and Smale designed a dynamic model to achieve the velocity consensus, and many researchers have proposed various models that uses the Cucker-Smale (CS) model [6]. Perea *et al.* [7] studied that the application of the CS model to the formation of spacecraft reduces the fuel consumption and the maximum distance between the spacecraft. By adding additional terms to the CS model, collision avoidance was performed in [8]. Similarly, adding additional terms to maintain the relative distance between neighbors called augmented Cucker-Smale model is developed in [9]. Kim and Peszek [10] analyzed modifying the augmented CS model for collision avoidance.

The aforementioned flocking models use a decentralized control method suitable for a large number of unmanned aerial vehicles (UAVs). The decentralized control structure is not hierarchical but all agents have equal authority. In flocking control, decentralized control has several advantages compared to centralized control. One is that the small amount of computation required for each agent could allow real-time onboard implementation. Another is that the scalability and flexability could allow the system to operate stably despite the malfunction or inability of some agents in a flock during the mission. The objective of this study is to make a suitable flocking flight in which many UAVs based on decentralized control successfully

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perform missions such as loitering around a certain point or tracking a target.

This study considers three types of guidjance law and fuses them to be applied to each UAV: vector field, Cucker-Smale model, and potential field. Fusing several guidance laws to make single guidance command would occur various problems. First of all, each guidance law could have a conflict with each other that can degrade the performance. For instance, the performance of the vector field guidance to follow the desired orbit could be degraded by the CS model. To mitigate this issue, we modify the parameters of the CS model adaptively. Besides, we propose the guidance scheme based on the weightings for three types of guidance law to solve the conflict. In [11], optimization was carried out to avoid collisions among agents with obstacles or walls in limited environments. Similarly, we carry out optimization to find suitable weightings for three guidance methods to improve the mission performance.

The rest of the paper is as follows. In Section II and III, we present the problem overview and guidance methods of hybrid flocking control and describe their features. In Section IV, the optimization technique for finding the best guidance weighting is described. Simulation results are given in Section V, followed by conclusions in Section VI.

II. Problem Overview

To form a flock, agents need to satisfy the Reynolds rule, the combination of three elements: cohesion, alignment, and separation. Cohesion means that agents fly together, alignment is responsible for velocity consensus between agents, and separation achieves that agents at the risk of collision move away from each other. One of the widely-used models that meet the Reynold rule and forms a flock is the Cucker-Smale model with the inter-agent bonding force term, which achieves velocity alignment and relative distance control among agents [9]. However, this model does not guarantee collision avoidance among agents explicitly. Even using a flocking model that theoretically guarantees collision avoidance, a more conservative approach might be required considering the constraints of small fixed-wing UAV and noise and delay in the actual flight environment. For this, we add additional repulsive force among agents for collision avoidance using the potential field method.

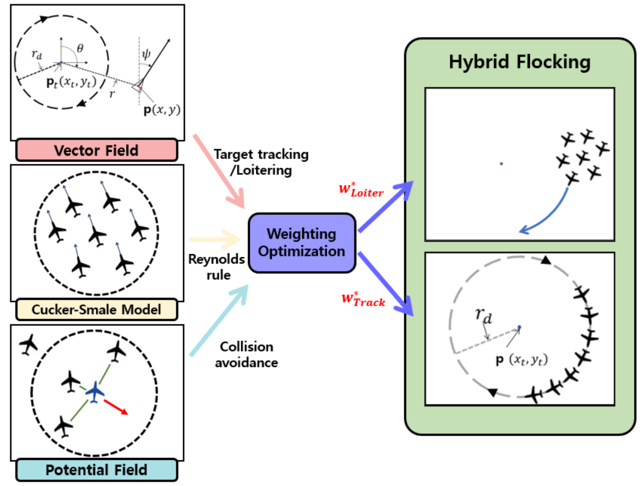

Fig. 1: Hybrid flocking guidance concept.

The significance of the flocking is not just in the reaching consensus itself but in that it functions as a preparatory stage for the cooperation. In this regard, the conventional flocking algorithm should be implemented together with the mission-specific guidance approach such as the vector field method for target tracking. Besides, waiting for the mission from the human operators, UAVs should be able to loiter around a given point. Based on the above reasons, this study proposes to use hybrid flocking guidance consisting of three types of guidance approaches as illustrated in Fig. 1.

In the decentralized hybrid flocking guidance, we assign the weightings to three types of guidance law. At this moment, it is important to assign appropriate weightings. If improper weighting is set for the mission, the performance of the mission could be poor or the mission can not be successfully executed. Therefore, in this study, the optimal weighting set for three types of guidance law is obtained by using social learning particle swarm optimization (SL-PSO) [12].

III. Hybrid Flocking Guidance

In this section, we discuss in detail the three guidance methods that constitute the hybrid flocking guidance.

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A. Vector Field Guidance

The dynamic model for guidance is given as:

$$\begin{aligned}\dot{x} &= v_{xy}\cos\psi,\\ \dot{y} &= v_{xy}\sin\psi,\\ \dot{z} &= v_z,\\ \dot{v}_{xy} &= U_1,\\ \dot{\psi} &= U_2,\\ \dot{v}_z &= U_3,\end{aligned} \tag{1}$$

where $\mathbf{p} = (x, y, z)^T$ is the position of the agent, $\mathbf{v} = (v_x, v_y, v_z)^T$ is the velocity of the agent, $\mathbf{U} = (U_1, U_2, U_3)^T$ is the control input, ψ is the heading angle, and $v_{xy} = \sqrt{v_x^2 + v_y^2}$. The control law is given as:

$$\begin{aligned}U_1 &= v_{spd} - v_{xy},\\ U_2 &= \psi_d - \psi,\\ U_3 &= z_t - z,\end{aligned} \tag{2}$$

where v_{spd} is the desired agent speed, ψ_d is the desired heading angle, and $\mathbf{p}_t = (x_t,\ y_t,\ z_t)^T$ is a loitering or target point. The guidance command in an inertial frame is derived as:

$$\begin{aligned}\mathbf{u}^{VF} &= R_\psi \mathbf{U},\\ R_\psi &= \begin{bmatrix}\cos\psi & -\sin\psi & 0\\ \sin\psi & \cos\psi & 0\\ 0 & 0 & 1\end{bmatrix}.\end{aligned} \tag{3}$$

The vector field guidance is a well-known method for a path-following or target tracking guidance problem [13]. The problem of circular orbit standoff target tracking for fixed-wing UAVs is solved through the vector field guidance. Using the Lyapunov stability theory, the convergence of the vector field guidance to the circular orbit is derived. The following desired velocity is formulated as:

$$f(\mathbf{p}) = \frac{-v_{spd}}{r(r^2 + r_d^2)}\begin{bmatrix}x_r(r^2 - r_d^2) + y_r(2rr_d)\\ y_r(r^2 - r_d^2) - x_r(2rr_d)\end{bmatrix} \tag{4}$$

where $x_r = x - x_t, y_r = y - y_t$, $r = \sqrt{x_r^2 + y_r^2}$ is the distance between the target and the agent, and r_d is the desired distance between the target and the agent. The desired heading from the above vector field can be computed as:

$$\psi_d^{circle} = \text{atan2}(f(2), f(1)). \tag{5}$$

Ellipse orbit vector field guidance can be generated by transforming the above circular vector field guidance. The vector field guidance of the ellipse orbit is required in order to reflect uncertainty in the target estimation problem [14]. The vector field guidance of the ellipse orbit can be obtained through the modification of the Eq. (4). It is generated by the linear transformation $\mathbf{p}' = C^{-1}\mathbf{p}$ where C is a nonsingular matrix and the following vector field guidance as:

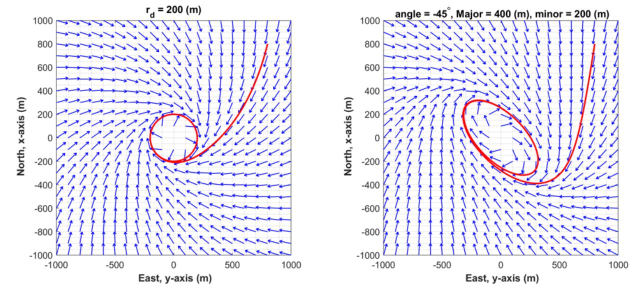

Fig. 2: Vector field directions and agent trajectory.

$$f'(\mathbf{p}) = C \cdot f(C^{-1}\mathbf{p}). \tag{6}$$

The matrix C can be obtained from a rotation matrix and a matrix that has diagonal components as a length of the ellipse's major and minor axis. The stability of (4) and (6) is proved in [14]. The desired heading of this ellipse vector field can be computed as:

$$\psi_d^{ellipse} = \text{atan2}(f'(2), f'(1)). \tag{7}$$

Figure 2 shows an example of the forces to particles in the circular and ellipse vector field. At these examples, the circular orbit r_d is $200(m)$ and linear transformation matrix for the ellipse orbit is

$$C = \begin{bmatrix}\cos(-45°) & \sin(-45°)\\ -\sin(-45°) & \cos(-45°)\end{bmatrix} \cdot \begin{bmatrix}400 & 0\\ 0 & 200\end{bmatrix} \tag{8}$$

where the rotation angle is $-45°$ and lengths of the major and minor axis are $400(m)$ and $200(m)$, respectively. Note that the rotation direction is counter-clockwise. Both bold red lines represent one of an agent's trajectory in vector fields.

B. Cucker-Smale Model

This study uses the augmented Cucker-Smale (CS) model to align the velocity and achieve the cohesion and separation [9], given as:

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$$
\begin{aligned}
\dot{\mathbf{p}}_i &= \mathbf{v}_i, \quad i \in \{1,2,\cdots,N\} \\
\dot{\mathbf{v}}_i &= \mathbf{u}_i^{CS} + \mathbf{u}_i^{Inter}, \\
\mathbf{u}_i^{CS} &= \frac{\lambda}{N}\sum_{j=1}^{N}\phi_1(\mathbf{p}_{ij})\big(\mathbf{v}_j - \mathbf{v}_i\big), \\
\mathbf{u}_i^{Inter} &= \frac{\sigma}{N}\sum_{j=1}^{N}\big(\phi_2(\mathbf{p}_{ij},\mathbf{v}_{ij}) + \phi_3(\mathbf{p}_{ij})\big) \\
&\qquad \cdot\big(\mathbf{p}_j - \mathbf{p}_i\big),
\end{aligned} \tag{9}
$$

where N is the number of agents, $\lambda, \sigma > 0$, $\mathbf{p}_{ij} = \mathbf{p}_i - \mathbf{p}_j, \mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$, and $\mathbf{p}_i, \mathbf{v}_i \in R^3$ are the position and velocity of the *i*-th agent, respectively. In Eq. (9), $\mathbf{u}_i^{CS}$ is the term to achieve the velocity consensus and $\mathbf{u}_i^{Inter}$ is the force acting in the relative distance direction to keep a desired distance between agents. The ϕ functions of each term are

$$
\begin{aligned}
\phi_1(\mathbf{p}_{ij}) &= 1/(1 + |\mathbf{p}_{ij}|^2)^{\beta}, \\
\phi_2(\mathbf{p}_{ij}, \mathbf{v}_{ij}) &= \frac{K_1}{2|\mathbf{p}_{ij}|^2}\langle \mathbf{v}_{ij}, \mathbf{p}_{ij}\rangle, \\
\phi_3(\mathbf{p}_{ij}) &= \frac{K_2}{2|\mathbf{p}_{ij}|}(|\mathbf{p}_{ij}| - 2R),
\end{aligned} \tag{10}
$$

where $\beta \geq 0, K_1, K_2 > 0, \langle\cdot,\cdot\rangle$ is the inner product, and $2R$ is the desired distance between agents. The constant β that represents the decay rate related to the relative distance determines if the agents can achieve the velocity consensus. Velocity consensus occurs when $\beta < 1/2$, and it occurs conditionally depending on the agents' initial states when $\beta \geq 1/2$ [6]. For analysis of the flocking system, Eq. (9) is decomposed into macroscopic (average) and microscopic (fluctuation) as:

$$
\mathbf{p}_c = \frac{1}{N}\sum_{i=1}^{N}\mathbf{p}_i, \quad \mathbf{v}_c = \frac{1}{N}\sum_{i=1}^{N}\mathbf{v}_i, \tag{11}
$$

$$
\hat{\mathbf{p}}_i = \mathbf{p}_i - \mathbf{p}_c, \quad \hat{\mathbf{v}}_i = \mathbf{v}_i - \mathbf{v}_c. \tag{12}
$$

Then, the macroscopic system (11) and microscopic system (12) dynamics are followed as:

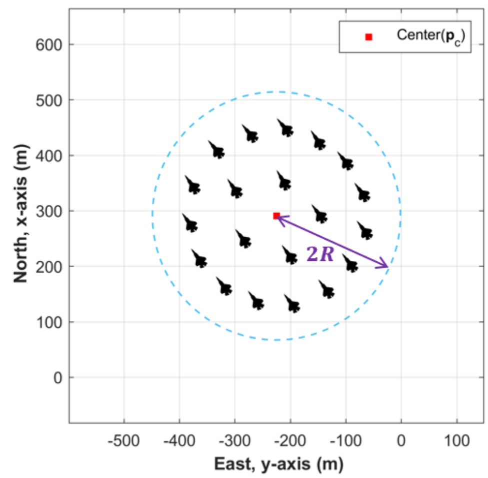

Fig. 3: Flocking configuration ($N = 20$).

$$
\begin{aligned}
\dot{\mathbf{p}}_c &= \mathbf{v}_c, \\
\dot{\mathbf{v}}_c &= \frac{\lambda}{N^2}\sum_{i=1}^{N}\sum_{j=1}^{N}\phi_1(\mathbf{p}_{ij})(\mathbf{v}_j - \mathbf{v}_i) + \\
&\quad \frac{\sigma}{N^2}\sum_{i=1}^{N}\sum_{j=1}^{N}\big(\phi_2(\mathbf{p}_{ij},\mathbf{v}_{ij}) + \phi_3(\mathbf{p}_{ij})\big) \\
&\qquad \cdot\big(\mathbf{p}_j - \mathbf{p}_i\big) \\
&= 0.
\end{aligned} \tag{13}
$$

and

$$
\begin{aligned}
\dot{\hat{\mathbf{p}}}_i &= \hat{\mathbf{v}}_i, \quad i \in \{1,2,\cdots,N\} \\
\dot{\hat{\mathbf{v}}}_i &= \hat{\mathbf{u}}_i^{CS} + \hat{\mathbf{u}}_i^{Inter}, \\
\hat{\mathbf{u}}_i^{CS} &= \frac{\lambda}{N}\sum_{j=1}^{N}\phi_1(\hat{\mathbf{p}}_{ij})\big(\hat{\mathbf{v}}_j - \hat{\mathbf{v}}_i\big), \\
\hat{\mathbf{u}}_i^{Inter} &= \frac{\sigma}{N}\sum_{j=1}^{N}\big(\phi_2(\hat{\mathbf{p}}_{ij},\hat{\mathbf{v}}_{ij}) + \phi_3(\hat{\mathbf{p}}_{ij})\big) \\
&\qquad \cdot\big(\hat{\mathbf{p}}_j - \hat{\mathbf{p}}_i\big),
\end{aligned} \tag{14}
$$

where $\hat{\mathbf{p}}_{ij} = \hat{\mathbf{p}}_i - \hat{\mathbf{p}}_j$ and $\hat{\mathbf{v}}_{ij} = \hat{\mathbf{v}}_i - \hat{\mathbf{v}}_j$. Note that $\mathbf{v}_c$ is constant and the same as initial $\mathbf{v}_c(0)$.

To prove the stability of the flocking system, consider the Lyapunov function as:

$$
V = \frac{1}{2}\sum_{i=1}^{N}|\hat{\mathbf{v}}_i|^2 + \frac{\sigma K_2}{8N}\sum_{i,j=1}^{N}\big(|\hat{\mathbf{p}}_{ij}| - 2R\big)^2. \tag{15}
$$

This Lyapunov function consists of terms to achieve the velocity consensus and the desired

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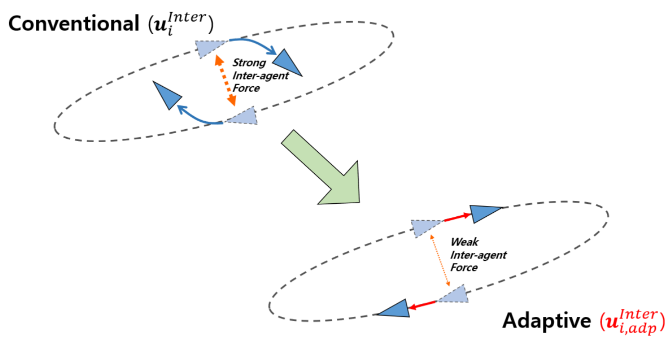

Fig. 4: Command change by adaptive control.

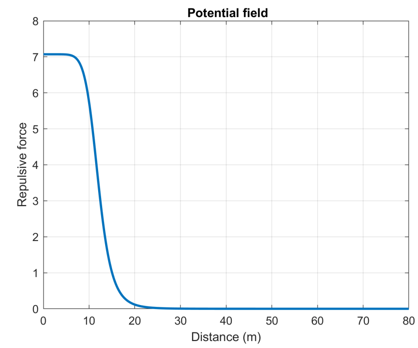

Fig. 5: Potential field for collision avoidance.

distances between agents. The time derivative of V is

$$\begin{aligned}\frac{dV}{dt} &= -\frac{\lambda}{2N}\sum_{i=1}^{N}\sum_{j=1}^{N}\phi_1(\hat{\mathbf{p}}_{ij})|\hat{\mathbf{v}}_{ij}|^2 \\ &\quad -\frac{\sigma K_1}{4N}\sum_{i=1}^{N}\sum_{j=1}^{N}\frac{\langle\hat{\mathbf{v}}_{ij},\hat{\mathbf{p}}_{ij}\rangle^2}{|\hat{\mathbf{p}}_{ij}|^2}.\end{aligned} \tag{16}$$

$\dot{V}$ is zero when $\hat{\mathbf{v}}_i = \hat{\mathbf{v}}_j,\ \forall i,j \in \{1,2,\cdots,N\}$. This means that all agents have the same velocity as the average velocity. Therefore, the magnitude of $\hat{\mathbf{v}}_i$ defined as the fluctuation from the average velocity becomes zero. This is the desired result that achieves the velocity consensus. It is worthwhile noting Eq. (13) that the aligned velocity is initial average velocity as $\dot{\mathbf{v}}_c = 0$.

Given the augmented CS model (9), the relative desired distance between the two agents is $2R$. However, when the number of agents exceeds the dimension of the physical space, it is impossible to satisfy the desired distance among all agents. Instead, in the convergence configuration, the spatial shape has a certain bound. From proposition 3.4 in [10], the bounds of the position in the converged state are

$$\begin{aligned}&\lim\sup_{t\to\infty}|\hat{\mathbf{p}}_i| \le 2R, \\ &\lim\sup_{t\to\infty}|\hat{\mathbf{p}}_i - \hat{\mathbf{p}}_j| \le 4R.\end{aligned} \tag{17}$$

In the convergence configuration, as illustrated in Fig. 3, the agents are distributed in a circle with a radius of $2R$ from the center of the agents and the relative distance between the agents is bounded by $4R$.

Applying multiple guidance commands at the same time can cause problems as mentioned earlier. In this work, the vector field guidance performance could be degraded by the augmented CS model. In other words, although many agents fly along the given orbit, the inter-agent force of the augmented CS model may lead to abnormal flight (see conventional in Fig. 4). To overcome this problem, the guidance command $\mathbf{u}_i^{Inter}$, which is intended to maintain the desired distance in the augmented CS model, is adaptively applied. It is implemented by changing the original constant σ in Eq. (9) into a function of the angle between the agents' velocity as:

$$\begin{aligned}\mathbf{u}_{i,adp}^{Inter} &= \frac{1}{N}\sum_{j=1}^{N}\sigma_{ij}\big(\phi_2(\mathbf{p}_{ij},\mathbf{v}_{ij}) + \phi_3(\mathbf{p}_{ij})\big) \\ &\quad \cdot(\mathbf{p}_j - \mathbf{p}_i), \\ \sigma_{ij} &= \sigma_1 \cdot \exp\Big(-\Big(\frac{\theta_{ij}}{\sigma_2}\Big)^2\Big),\end{aligned} \tag{18}$$

where $\theta_{ij} = \cos^{-1}\dfrac{\langle\mathbf{v}_i,\mathbf{v}_j\rangle}{|\mathbf{v}_i||\mathbf{v}_j|}$ and $\sigma_1,\sigma_2 > 0$. This σ function decreases when the velocity angle θ_{ij} increases (i.e., when velocities are the opposite direction as shown in adaptive in Fig. 4).

C. Potential Field Guidance

One of the most important factors in operating multiple UAVs is preventing collisions between the UAVs. In the augmented CS model, although $\mathbf{u}_{i,adp}^{Inter}$ can generate the repulsive force between agents, it may be insufficient to avoid the collision since the guidance command is divided by the number of agents and decreases with the angle between the agents' velocity. In other words, the CS model does not guarantee collision avoidance. Therefore, additional repulsion is required between agents with high-risk collision. In this study, potential field guidance is used for collision avoidance. A

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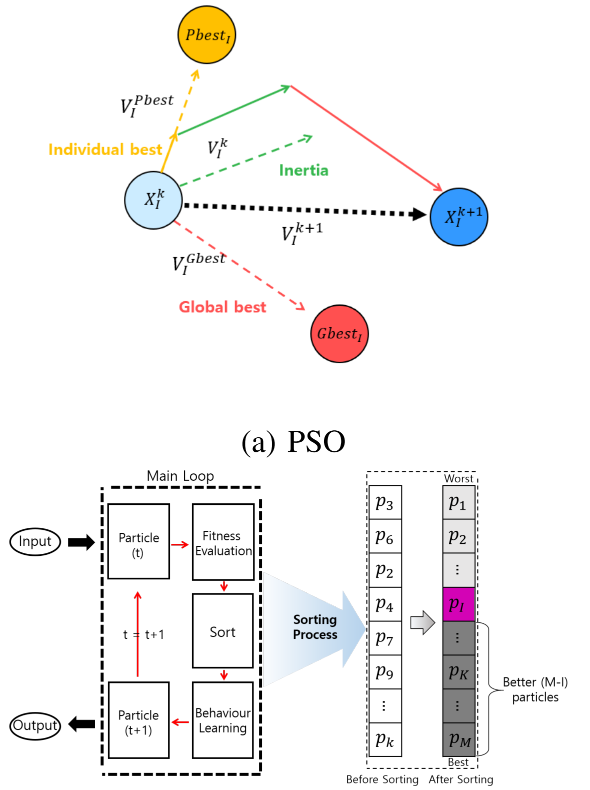

(a) PSO

(b) SL-PSO

Fig. 6: Concept of the PSO-based optimization algorithms.

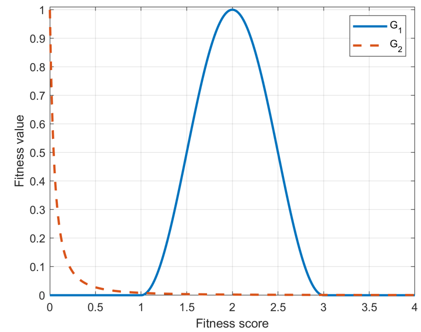

Fig. 7: Fitness functions $(G_1(x, 2, 1), G_2(x, 0.1))$.

bell-shaped function is introduced for generating an additional repulsion as the distance between agents gets closer. The potential field guidance is defined as:

$$\dot{\mathbf{v}}_i = \mathbf{u}_i^{PF} = \sum_{j=1}^{N} \frac{\alpha}{1 + \left| \frac{|\mathbf{p}_{ij}| - c}{a} \right|^{2b}} \cdot \frac{\mathbf{p}_i - \mathbf{p}_j}{|\mathbf{p}_{ij}|}, \quad (19)$$

where $\alpha, a, b > 0$ and c are constant parameters. Figure 5 illustrates the potential field used for collision avoidance in this study.

IV. Weighting Optimization

The social learning particle swarm optimization (SL-PSO) [12] is employed to optimize the weighting of three guidance commands. It adds a sociological factor to conventional particle swarm optimization (PSO). Each particle stochastically learns the information of other particles having a better cost than itself. We briefly introduce the concept here.

Figure 6(a) shows the concept of PSO. The PSO algorithm is a search method using spatially-distributed particles to find the optimal cost in a given domain space where the particles acquire the cost for their position at every iteration. Each particle obtains its cost and continues to search in the direction of the weighted sum of the best value of itself (called as *Pbest*), the global best among all particles (*Gbest*), and its current motion direction until most of the particles converge to the optimal cost.

Although PSO can find the optimal solution effectively, it is easy to fall into local minima due to the lack of global search capability and the large memory usage due to the memory of past particle information. In order to address this problem, SL-PSO is adopted in this study among various variations of PSO. The difference between SL-PSO and PSO is a learning process from other particles. In PSO, all particles learn from the global best value at every iteration. Whereas in SL-PSO, all M particles are sorted according to the cost and each particle stochastically learns from the randomly-chosen particle which has a better cost than itself. In other words, the I-th ranked particle randomly learns from one of the $(M - I)$ particles that have better cost as illustrated in Fig. 6(b). This can be interpreted as the addition of the sociological learning theory to the PSO method. Through this process, SL-PSO converges towards the optimal solution better. Compared to the PSO algorithm, SL-PSO has a number of benefits. First, it is a global search technique and it can easily escape from the local minima. Second, the memory usage is low as there is no need to memorize

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the past cost value. Lastly, parameter tuning is not necessary which makes this algorithm easy to use. The overall operating sequence of SL-PSO is depicted in Fig. 6(b).

We need to design a cost function to determine the optimal weighting for our mission. Several indicators are introduced for the alignment, cohesion, speed following, collision avoidance, and vector field guidance. The fitness values for alignment and cohesion are defined as:

$$
\begin{aligned}
F^{Align} &= \frac{1}{T}\frac{1}{N(N-1)}\int_0^T \sum_{i=1}^{N}\sum_{j=1,i\neq i}^{N} \frac{\langle \mathbf{v}_i, \mathbf{v}_j\rangle}{|\mathbf{v}_i||\mathbf{v}_j|}dt,\\
F^{Coh} &= \frac{1}{T}\frac{1}{N}\int_0^T \sum_{i=1}^{N} H(2R - |\mathbf{p}_c - \mathbf{p}_i|)dt,\\
H(x) &= \begin{cases} 1, & \text{if } x > 0,\\ 0, & \text{otherwise,}\end{cases}
\end{aligned}
\tag{20}
$$

where T is the simulation time and H is the unit step function. Looking at each fitness values, F^{Align} is large when the angle of the velocity between the agents is small. F^{Coh} is large when the relative distance between the agent and the center is smaller than $2R$. The fitness values for speed following and collision avoidance are defined as:

$$
\begin{aligned}
F^{Spd} &= G_1(S^{Spd}, v_{spd}, v_{tor}),\\
F^{PF} &= G_2(S^{PF}, a_{col}),
\end{aligned}
\tag{21}
$$

where G_1 and G_2 are functions for evaluating the fitness value, S^{Spd} and S^{PF} are fitness scores of the speed following and the collision avoidance, v_{tor} is the tolerance value for speed, and a_{col} is the collision risk gain. The fitness functions G_1, G_2 are defined as:

$$
\begin{aligned}
G_1(x, x_0, d) &= \begin{cases} \frac{1}{2}(1+\cos\frac{\pi}{d}(x-x_0)), & \\ \quad \text{if } x_0 - d \le x \le x_0 + d, & \\ 0, & \\ \quad \text{otherwise,} & \end{cases}\\
G_2(x,\gamma) &= \frac{\gamma^2}{(x+\gamma)^2}.
\end{aligned}
\tag{22}
$$

The fitness functions are illustrated in Fig. 7. By Eq. (22), F^{Spd} and F^{PF} have large values as S^{Spd} is equal to v_{spd} and S^{PF} is small, respectively. The fitness scores S^{Spd} and S^{PF} are defined as:

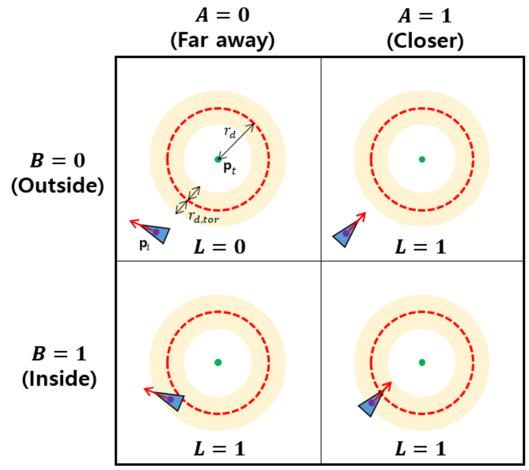

Fig. 8: Illustration for the fitness score of the vector field guidance.

$$
\begin{aligned}
S^{Spd} &= \frac{1}{T}\frac{1}{N}\int_0^T \sum_{i=1}^{N}|\mathbf{v}_i|dt,\\
S^{PF} &= \\
&\frac{1}{T}\frac{1}{N(N-1)}\int_0^T \sum_{i=1}^{N}\sum_{j=1,j\neq i}^{N} H(r_{col} - |\mathbf{p}_{ij}|)dt,
\end{aligned}
\tag{23}
$$

where r_{col} is the collision distance. S^{Spd} gets a score according to the agents' speed and S^{PF} gets a score if the relative distance between agents is less than r_{col}.

There are two fitness values defined for vector field guidance. One is needed when agents are turning around the target for loitering. The other is needed when they are required to perform a target tracking and follow the given orbit precisely. These are defined as:

$$
\begin{aligned}
F^{VF}_{Loiter} &= S^{VF},\\
F^{VF}_{Track} &= G_1(S^{VF}, r_{track}, r_{tor}),
\end{aligned}
\tag{24}
$$

where S^{VF} is the fitness score of the vector field guidance, r_{track} is the desired tracking fitness score and r_{tor} is the tolerance value for tracking fitness score. The loiter fitness value is the S^{VF}. Meanwhile, since it is important to follow the orbit correctly, the tracking fitness value uses the fitness function G_1. The fitness score S^{VF} is defined as:

$$
S^{VF} = \frac{1}{T}\frac{1}{N}\int_0^T \sum_{i=1}^{N} L(A,B)dt. \tag{25}
$$

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$L(A,B)$ is a function indicating if agents follow the given orbit. It is defined as:

$$\begin{aligned} L(A,B) &= \begin{cases} 0, & \text{if } A = B = 0\ , \\ 1, & \text{otherwise}, \end{cases} \\ A &= H\big(-\frac{d}{dt}\big||\mathbf{p}_t - \mathbf{p}_i| - r_d\big|\big), \\ B &= H\big(r_{d,tor} - \big||\mathbf{p}_t - \mathbf{p}_i| - r_d\big|\big), \end{aligned} \tag{26}$$

where $r_{d,tor}$ is the tolerance value for tracking error. $L(A,B)$ is zero when both A and B are zero. Otherwise, $L(A,B)$ is one. When the distance between the agent and the target ($|\mathbf{p}_t - \mathbf{p}_i|$) gets closer to the desired distance (r_d), A is one. When the difference between $|\mathbf{p}_t - \mathbf{p}_i|$ and r_d is less $r_{d,tor}$, B is one as illustrated in Fig. 8.

The integrated cost function that takes into account the fitness values defined in Eqs. (20)–(26) is defined as:

$$F_{Loiter} = F^{Align} \cdot F^{Coh} \cdot F^{Spd} \cdot F^{PF} \cdot F^{VF}_{Loiter}, \tag{27}$$

$$F_{Track} = F^{Align} \cdot F^{Coh} \cdot F^{Spd} \cdot F^{PF} \cdot F^{VF}_{Track}. \tag{28}$$

Accordingly, the weighting optimization problem can be expressed as the following maximization:

$$\mathbf{w}^*_{Loiter} = \arg\max F_{Loiter}, \tag{29}$$

$$\mathbf{w}^*_{Track} = \arg\max F_{Track}, \tag{30}$$

where $\mathbf{w}^*_{Loiter}$ and $\mathbf{w}^*_{Track}$ are the three-dimensional vector with normalizing to 1-norm. The optimized weight is applied to each agent as follows:

$$\mathbf{u}_i = w_1^* \mathbf{u}_i^{VF} + w_2^*(\mathbf{u}_i^{CS} + \mathbf{u}_i^{Inter}) + w_3^* \mathbf{u}_i^{PF}, \tag{31}$$

where $\mathbf{w}^* = (w_1^*, w_2^*, w_3^*)$ is optimized weight. This command will result in the best performance for the given mission.

V. Simulation Results

In this section, numerical simulation results of the decentralized hybrid flocking guidance with the optimized weighting set are presented and analyzed. All agents' initial position and velocity are randomized for Monte-Carlo simulations with 50 trials. Parameters used in the simulation are listed in Table I. In the table, L_{bound} is used for the initial deployment boundary of agents which bounds them within a square whose edge length is L_{bound}.

TABLE I: Simulation parameters

Parameter	**Value**	**unit**
Desired speed, v_{spd}	20	m/s
Number of agents, N	20	
Sample time, Δt	0.5	s
Terminal time, T	300	s
Desired relative distance, $2R$	75	m
Communication decay rate, β	0.25	
Alignment strength, λ	1	
Bonding force strength, $(\sigma 1, \sigma 2)$	(1, π)	
Bonding gain, (K_1, K_2)	(0.32, 0.0256)	
Potential strength, α	$5\sqrt{2}$	
Bell function constants, (a, b, c)	(12, 4, 0)	
Max/Min speed, (v_{max}, v_{min})	(25, 15)	m/s
Maximum acceleration, $\dot{v}_{max}$	2	m/s^2
Collision risk distance, r_{col}	20	m
Collision risk gain, a_{col}	0.1	
Tolerance for speed, v_{tor}	2	m/s
Tolerance for desired radius, $r_{d,tor}$	45	m
Desired tracking fitness, r_{track}	1	
Tolerance for tracking fitness, r_{tor}	0.1	
Initial location bound, L_{bound}	250	m

TABLE II: The optimization results

	Vector field	CS model	Potential field
$\mathbf{w}^*_{Loiter}$	0.0318	0.4849	0.4833
$\mathbf{w}^*_{Track}$	0.3038	0.1403	0.5559

In this study, we propose the adaptive gain (σ_{ij} as in Eq. (18)) of the augmented CS model to form the flock and follow the given orbit. To verify that the proposed hybrid guidance works effectively, we compare the case with and without adaptive guidance. Figure 9 shows the trajectories for the agents and the circular/ellipse orbits to be followed. Figure 9(a) shows the trajectories when using the conventional augmented CS model and Fig. 9(b) shows the trajectories when applying the proposed adaptive CS model of (18). If we compare two trajectories, Fig. 9(b) follows the given orbit better. Besides, the F^{VF}_{Loiter} value using the conventional model and the adaptive model are 0.4019 and 0.6018, respectively. This means that the proposed adaptive augmented CS model improves the mission performance.

Next, the optimized weighting set for missions are analyzed. Table II shows the optimized weighting set by the cost function of Eqs. (27) and (28). The optimized weight $\mathbf{w}^*_{Loiter}$ in terms of F_{Loiter} has a small vector field guidance weight about 0.03, while the weight of the CS model

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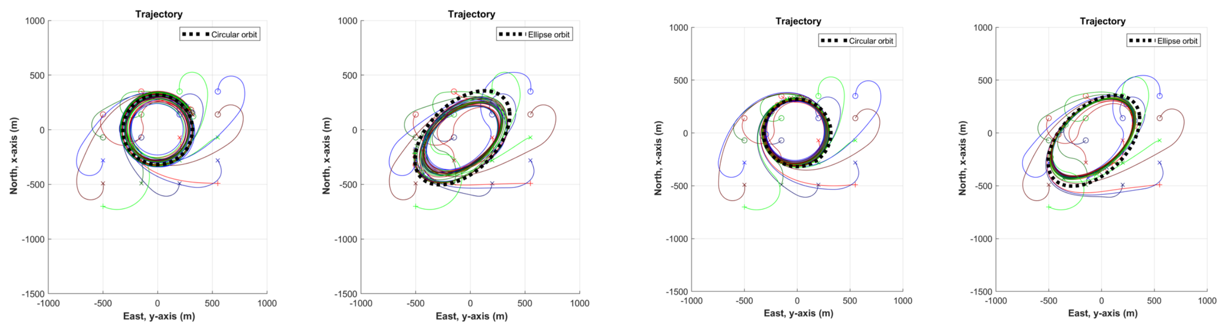

(a) Conventional (b) Adaptive

Fig. 9: Trajectory comparison.

is large about 0.48. This means that the vector field guidance should be small for turning around the target point effectively in terms of the hybrid flocking. Furthermore, due to the large CS model weight, agents fly together within a closely packed area.

The optimized weight of $\mathbf{w}^*_{Track}$ in terms of F_{Track} has a large vector field guidance weight about 0.3, while the weight of the CS model is small about 0.14. Accurately tracking the given orbit while forming the flock, the cost of F^{VF}_{Track} should be large. If the agents do not follow the given orbit exactly, the cost of F^{VF}_{Track} decreases drastically depending on the r_{track} and r_{tor}. To be responsible for this property, r_{track} and r_{tor} are 1 and 0.1, respectively. In $\mathbf{w}^*_{Loiter}$, the relationship between vector field guidance and CS model is not explicitly relevant. On the other hand, $\mathbf{w}^*_{Track}$ shows a certain relationship between vector field guidance and CS model, because the only appropriate weights of $\mathbf{w}^*_{Track}$ can make agents follow given orbit while flying together.

The optimized weight for potential field guidance is large in both missions. Since the desired relative distance is small in this work ($2R = 75m$), collision risk between the agents is high until forming the flock. Besides, since the fitness function G_2, illustrated in Fig. 7, decreases sharply even with a small collision risk, it makes the weight of the potential field guidance large.

Figure 10 shows the results of flocking in arbitrary initial states using the weight of the $\mathbf{w}^*_{Loiter}$. Figure 10(a) shows the agents' trajectories. The agents are closely gathered while turning around the target. This is due to the small weight for vector field guidance and the large weight for CS model in the $\mathbf{w}^*_{Loiter}$. Looking at the velocity on the xy-plane in Fig. 10(b), which shows the velocity of the agents, all agents have the similar velocity. This shows that all agents are flying in the same direction as illustrated in Fig. 3. Figure 10(c), representing the relative distance, shows that no collision occurred.

Figure 11 shows the results of flocking using the weight of the $\mathbf{w}^*_{Track}$. Figure 11(a) shows the agents' trajectories and given orbits. Unlike Fig. 10(a), the agents follow the given orbit precisely. Besides, the flocking configuration of Fig. 10(a) is dense within a certain radius, while Fig. 11(a) shows that the flocking configurations are aligned along the orbit. This behavior can be identified by the agents' velocity as well. Figure 11(b) shows that the agents have the similar velocity over time with other agents. Figure 11(c) shows that no collision between agents.

VI. Conclusions and Future Work

In this study, the decentralized hybrid flocking guidance approach was proposed based on vector field guidance, Cucker-Smale models, and potential field guidance. The augmented Cucker-Smale model is modified to adaptively perform the flocking and target tracking mission at the same time. Using the social learning particle swarm optimization technique, we found that the optimized weighting set effectively perform the flocking mission. We will conduct the flocking experiment utilizing the identified optimal weighting set for each mission.

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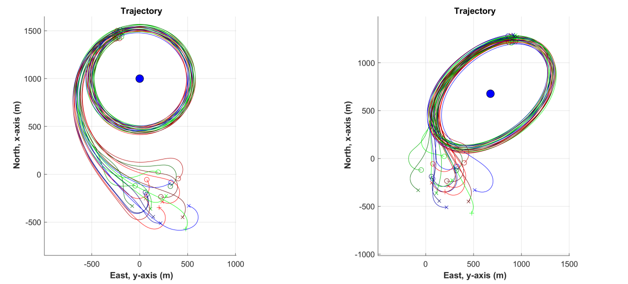
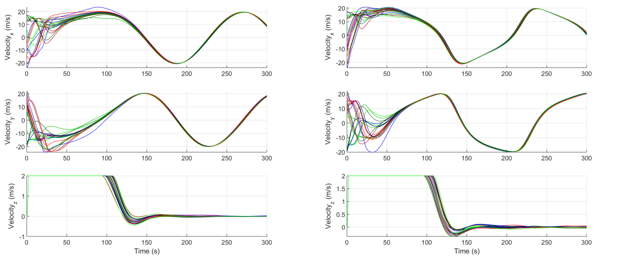
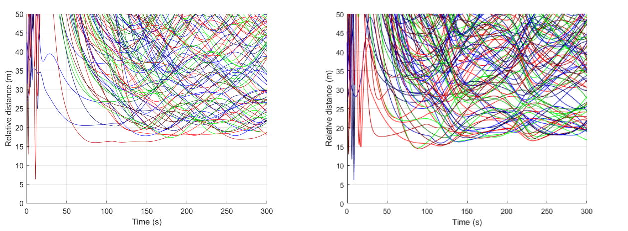


(a) Trajectory

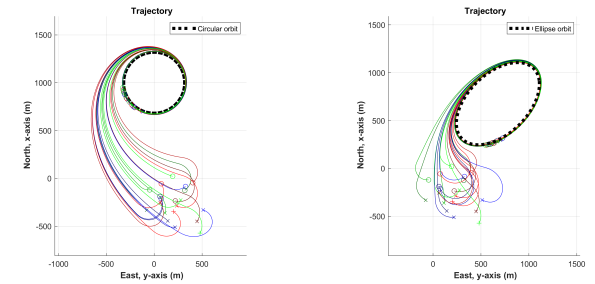


(b) Velocity

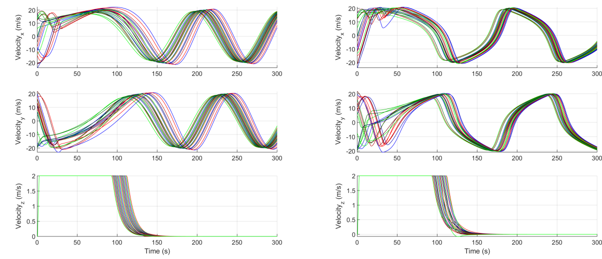


(c) Relative distance

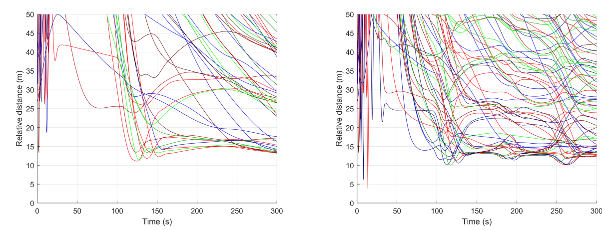
Fig. 10: States when using $\mathbf{w}^*_{Loiter}$.



(a) Trajectory



(b) Velocity



(c) Relative distance

Fig. 11: States when using $\mathbf{w}^*_{Track}$.

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